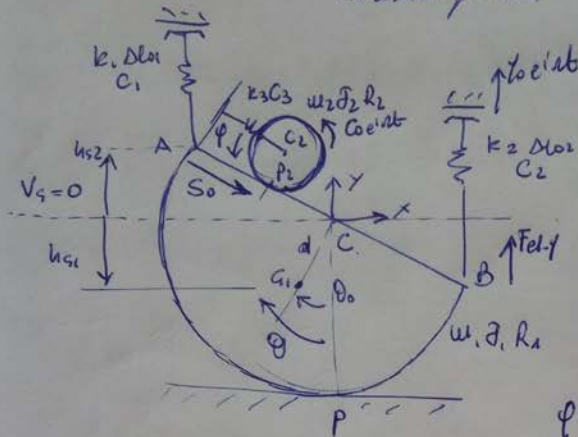


- Standard problem -



2 def. $\underline{x} = \begin{Bmatrix} \theta \\ \phi \end{Bmatrix}$ $x_0 = \begin{Bmatrix} \theta_0 \\ \phi_0 \end{Bmatrix}$

$$T = \frac{1}{2} m_1 \dot{x}_{G1}^2 + \frac{1}{2} m_1 \dot{y}_{G1}^2 + \frac{1}{2} I_G \dot{\theta}^2 + \frac{1}{2} m_2 \dot{x}_{G2}^2 + \frac{1}{2} m_2 \dot{y}_{G2}^2 + \frac{1}{2} I_A \dot{\phi}^2$$

ϕ is the relative rotation with respect to the 1st body.

$$D = \frac{1}{2} \sum_{i=1}^2 C_i \dot{\Delta}_i^2, \quad V_{EL} = \frac{1}{2} \sum_{i=1}^2 k_i \Delta_i^2 \quad \text{respect the 1st body.}$$

$$SW = \text{Coef}^{\text{ist}} J_A^P + F_{\text{el-y}} J_B^P \quad ; \quad F_{\text{el-y}} = (k_2 \text{ in } C_2) \% \text{ ein}$$

taking into account the rolling without sliding condition $x_c = (10 - 80)R$.

$$X_{G1} = X_c + X_{G1c} = (\theta - \theta_0) R_1 - d \sin \theta$$

$$Y_{G1} = -d \cos \theta$$

$$x_A = (\theta - \theta_0) R_1 - R_1 \cos \theta$$

$$Y_{\Delta} = R_1 \sin \theta$$

$$x_{C2} = (\theta - \theta_0) R_1 - (R_1 - S_0 + R_2 \varphi) \cos \theta + R_2 \sin \theta$$

$$V_{C2} = (R_1 - S_0 + R_2 \phi) \sin \theta + R_2 \cos \theta$$

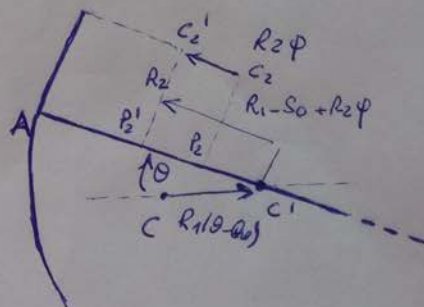
$$\mathcal{I}_B = -R_1 \sin \theta$$

$$\Delta L_1 = -\gamma_A + \gamma_{A\phi}$$

$$\Delta L_2 = -\gamma_B + \gamma_{B\phi}$$

$$\Delta L_3 = -R_2 \phi$$

$$h_{s1} = \gamma_{s1} \quad , \quad h_{s2} = \gamma_{s2}$$



$$\frac{\partial x_{A1}}{\partial \theta} = R_1 - d \cos \theta, \quad \frac{\partial y_{A1}}{\partial \theta} = d \sin \theta$$

$$\frac{\partial x_{A2}}{\partial \theta} = R_1 + (R_1 - S_0 + R_2 \phi) \sin \theta + R_2 \cos \theta, \quad \frac{\partial x_{A2}}{\partial \phi} = -R_2 \cos \theta$$

$$\frac{\partial y_{A2}}{\partial \theta} = (R_1 - S_0 + R_2 \phi) \cos \theta - R_2 \sin \theta, \quad \frac{\partial y_{A2}}{\partial \phi} = R_2 \sin \theta$$

$$\frac{\partial \Delta L_1}{\partial \theta} = -R_1 \cos \theta, \quad \frac{\partial \Delta L_2}{\partial \theta} = R_1 \cos \theta, \quad \frac{\partial Y_B}{\partial \theta} = -R_1 \cos \theta$$

$$\psi_A = \phi - \theta$$

$$[\Lambda_{A1}] = \begin{bmatrix} R_1 - d \cos \theta_0 & \phi \\ d \sin \theta_0 & \phi \\ 1 & \phi \\ R_1 + (R_1 - S_0) \sin \theta_0 + R_2 \cos \theta_0 & -R_2 \cos \theta_0 \\ (R_1 - S_0) \cos \theta_0 - R_2 \sin \theta_0 & R_2 \sin \theta_0 \\ -1 & 1 \end{bmatrix}$$

$$[\Lambda_{A2}] = \begin{bmatrix} -R_1 \cos \theta_0 & \phi \\ R_1 \cos \theta_0 & \phi \\ \phi & -R_2 \end{bmatrix}$$

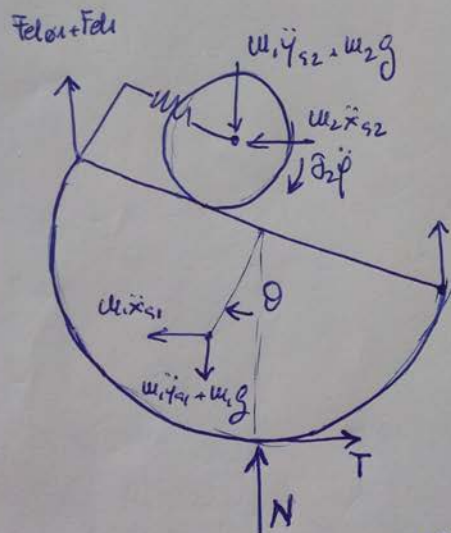
$$[\Lambda_B] = \begin{bmatrix} -1 & 1 \\ -R_1 \cos \theta_0 & \phi \end{bmatrix}$$

$$\frac{\partial^2 x_{A1}}{\partial \theta^2} = d \cos \theta, \quad \frac{\partial^2 \Delta L_1}{\partial \theta^2} = R_1 \sin \theta, \quad \frac{\partial^2 \Delta L_2}{\partial \theta^2} = -R_1 \sin \theta$$

$$\begin{cases} \frac{\partial^2 x_{A2}}{\partial \theta^2} = -(R_1 - S_0 + R_2 \phi) \sin \theta - R_2 \cos \theta \\ \frac{\partial^2 x_{A2}}{\partial \theta \partial \phi} = R_2 \cos \theta, \quad \frac{\partial^2 x_{A2}}{\partial \phi^2} = 0 \end{cases}$$

$$[k_{s1}] = m_1 g \begin{bmatrix} d \cos \theta_0 & \phi \\ \phi & \phi \end{bmatrix} \quad [k_{s2}] = m_2 g \begin{bmatrix} -(R_1 - S_0) \sin \theta_0 - R_2 \cos \theta_0 & R_1 \cos \theta_0 \\ R_2 \cos \theta_0 & \phi \end{bmatrix}$$

$$[k_{EL, B1}] = k_1 \Delta L_1 \begin{bmatrix} R_1 \sin \theta_0 & \phi \\ \phi & \phi \end{bmatrix} \quad [k_{EL, B2}] = k_2 \Delta L_2 \begin{bmatrix} -R_1 \sin \theta_0 & \phi \\ \phi & \phi \end{bmatrix}$$

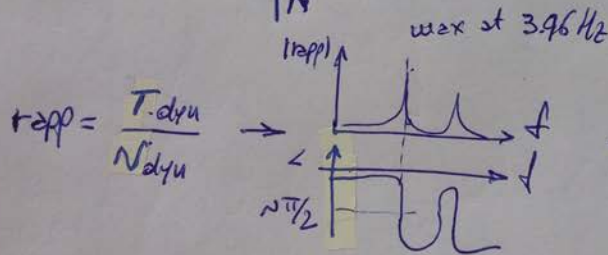


$$\sum F_x = 0 : T = m_1 \ddot{x}_{s1} + m_2 \ddot{x}_{s2} = T_{dyn}$$

$$\sum F_y = 0 : N = m_1 g + m_2 g + \underbrace{F_{EL1} + F_{EL2}}_{N_{sr}} + \underbrace{m_1 \ddot{y}_{s1} + m_2 \ddot{y}_{s2} - F_{EL1} - F_{EL2}}_{N_{dyn}}$$

$$F_{EL2} = (k_2 + i \Delta C_2) \Delta L_2 \quad (*)$$

$$\Delta L_2 = (y_0 + (R_1 \cos \theta_0) \theta)$$



As the relative phase between T_{dyn}/N_{dyn} is $\sim \pi/2$, when $|T_{dyn}|$ is max or min, $|N_{dyn}|$ is zero $\rightarrow$

$$\left. \frac{T_{dyn}}{y_0} \right|_{3.96} \cdot y_{0max} = N_{sr} f_s \rightarrow y_{0max} = \frac{N_{sr} f_s}{\left. T_{dyn}/y_0 \right|_{3.96 \text{ Hz}}}$$

(*) same result considering $\Delta L_2 = (R_1 \cos \theta_0) \theta$ and applying the equivalent force $F_{EL,y}$ together with F_{EL2}